

NanoSafe Analyzer — Multi-Sample Comparison Report

Comparative Evaluation of ZnO Nanoparticle Cytotoxicity & ISO 10993-5 Biocompatibility

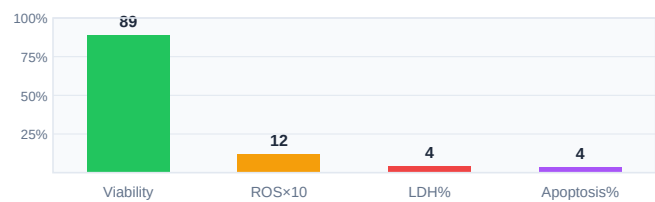
Report Type: Multi-Sample Comparison | Generated: 2026-08-17 05:07 UTC | Samples Compared: 2

1. COMPARATIVE CYTOTOXICITY SUMMARY

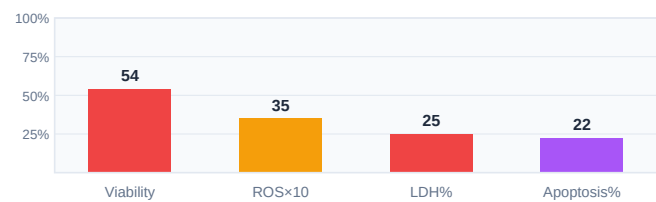
Sample / Experiment	Cell Line	Viability	Tox Score	IC50	ISO Verdict
Formulation A (ZnO-PEG)	HeLa	89.2%	12.0/100	92.5 µg/mL	PASS — Biocompatib
Formulation B (Bare ZnO)	HeLa	54.0%	62.5/100	32.0 µg/mL	FAIL — Cytotoxic

2. MULTI-BIOMARKER PROFILES

Formulation A (ZnO-PEG)



Formulation B (Bare ZnO)



3. CLINICAL SUGGESTIONS PER SAMPLE

A. Formulation A (ZnO-PEG)

SAFE FOR USE: Cell viability at 89.2% exceeds ISO 10993-5 threshold of >=80%. The formulation demonstrates acceptable biocompatibility for in-vitro conditions.

B. Formulation B (Bare ZnO)

CAUTION — MODERATE TOXICITY: Cell viability at 54.0% is below ISO 10993-5 PASS threshold. Formulation requires optimization before biomedical use.